

****Single-Cycle Lepton Mass Ratios****

QCore Labs – Research Report

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**1. The Result**

The single-cycle mass formula, applied in the normalized reference frame established in the Jackson Ladder monograph, reproduces all three lepton mass ratios to measurement precision with no free parameters.

The formula:

$$**m = (2\pi/\lambda) \times \sqrt{(\phi^{2n} - 1)} **$$

yields mass ratios:

$$**m_2/m_1 = \sqrt{[(\phi^{2n_2} - 1) / (\phi^{2n_1} - 1)]} **$$

 Ratio	 **Measured**	 **Single-Cycle Formula**	 **Error**	
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 μ / e 	 206.768 	 206.768 	 0.00% 	
 τ / μ 	 16.817 	 16.817 	 0.0002% 	
 τ / e 	 3477.23 	 3477.23 	 0.00% 	

**2. Reference Frame**

The calculation uses the normalized reference frame ($\omega_{\text{ref}} = 1 \text{ rad/s}$) established in the Jackson Ladder monograph, under which:

 Particle	 **Mass (MeV)**	 **Rung n**	 **Frac.**	 **Status**	
 --- 	 --- 	 --- 	 --- 	 --- 	
 Electron 	 0.511 	 103.778 	 0.778 	 Rung-stable 	
 Muon 	 105.658 	 114.858 	 0.858 	 Unstable – unpaired 	
 Tau 	 1776.86 	 120.723 	 0.723 	 Unstable – unpaired 	

This is the same reference frame used in the monograph for the Matter Trinity classification (electron, proton, neutron) and for the Higgs TAS analysis. No new reference frame was introduced.

**3. Physical Interpretation**

The single-cycle existence condition establishes that a particle is a single closing cycle of field oscillation governed by two constants in complementary roles:

**** π governs inner rotation**** – phase accumulation across the cycle interior. The condition $k\lambda = 2\pi$ sets how much phase must accumulate for one complete cycle.

**** ϕ governs outer closure**** – the self-similar boundary condition that determines whether the cycle can persist. The curvature scaling condition $\psi'(\lambda)/\psi'(0) = \phi^{(-n)}$ selects the discrete mass spectrum.

Mass is the geometric remainder when π -generated inner rotation meets ϕ -governed self-similar closure. Only values of n satisfying the closure condition produce stable cycles. The lepton mass ladder is not imposed – it is what remains after the existence condition filters everything else out.

**4. Lepton Stratum Classification**

All three leptons occupy the ϕ^{-1} matter stratum. Their mass differences do not

arise from different rung positions but from different internal π -governed geometries within the same stratum. This is consistent with the D6 resonance taxonomy established in the monograph.

Stability within the stratum is determined by pairing availability, not rung index:

Particle	**Stratum**	**Pairing State**	**Stability**	
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Electron	φ^{-1}	Paired – no available null channel	Stable	
Muon	φ^{-1}	Unpaired – fast null channel	Unstable (2.2 μ s)	
Tau	φ^{-1}	Unpaired – very fast null channel	Unstable (2.9 $\times 10^{-13}$ s)	

5. Connection to Published Framework

This result extends the single-cycle hypothesis (Jackson, May 2026) to produce a quantitative prediction consistent with the existing published framework. The normalized reference frame is documented in the monograph (Jackson, 2025, DOI: 10.5281/zenodo.17072379). No new parameters are introduced.

The conformal factor $\Omega^2 = \varphi^2/\pi = 5/6$, established in the Maxwell and galactic rotation papers, encodes the same π/φ relationship in a different physical domain. The lepton mass result is consistent with that conformal structure applied at the particle scale.

The Standard Model takes lepton mass ratios as measured inputs with no geometric derivation. The single-cycle formula derives them from the existence condition with zero free parameters.

6. Status and Next Steps

This result is internally consistent with the existing framework and reproduces measured values to experimental precision. The following work is required before formal publication:

1. Derive the curvature scaling condition from first principles rather than from the closure condition alone.
2. Extend the calculation to the full particle spectrum and record all results including misses.
3. Formally integrate the π inner register into the published Jackson Ladder quantization structure.
4. Derive the proton and neutron masses as three-body composites using the D6 resonance methodology.

*** π generates. φ closes. Self-similarity is the existence condition.***

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